

Poisson regression

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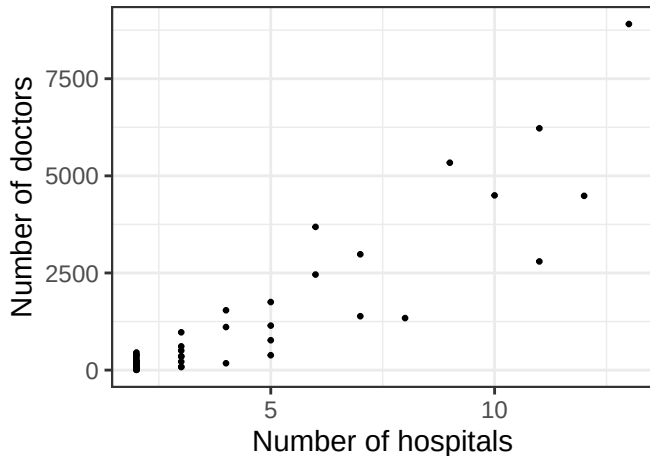
Count variables

Data: Data on medical facilities and doctors from a sample of 53 different counties in the US. Variables include:

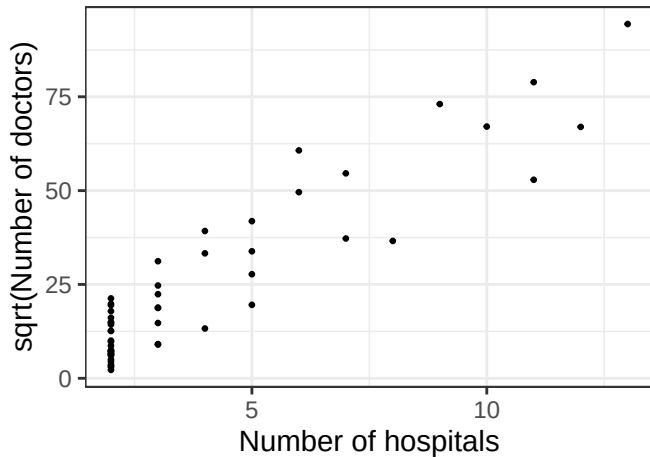
- ▶ MDs: the number of medical doctors in the county
- ▶ Hospitals: the number of hospitals in the county

Research question: Can we model the relationship between the number of hospitals and the number of doctors?

Plotting the data



Trying a transformation



Is a linear regression model appropriate now?

Limitation of transformations

Poisson regression

Fitting the Poisson regression model

```
m1 <- glm(MDs ~ Hospitals, data = CountyHealth,  
          family = poisson)  
summary(m1)$coefficients
```

##	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	5.1168958	0.009801022	522.0778	0
## Hospitals	0.3124424	0.001047882	298.1658	0

Interpreting the Poisson regression model

```
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Exponential dispersion models

Activity

Work on the activity (handout).